

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 October 2023 Worked Solutions

Oct 2023 paper rebuilt with the worked solution after each question.

11

paper questions

WMA11

Pure 1

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - eliteigcse.com

P1

PAST PAPER

WMA11/01 October 2023

Oct 2023 | 11 questions | 75 marks

11

questions

75

marks

Answers

worked solution
after each
question

Question 1

Differentiation

1. Given that

$$y = 5x^3 + \frac{3}{x^2} - 7x \quad x > 0$$

find, in simplest form,

(a) $\frac{dy}{dx}$ (3)

(b) $\frac{d^2y}{dx^2}$ (2)

(Total for Question 1 is 5 marks)

Worked Solution - Question 1

1. Rewrite the powers

$$y = 5x^3 + 3x^{-2} - 7x.$$

2. Differentiate once

$$\frac{dy}{dx} = 15x^2 - 6x^{-3} - 7 = 15x^2 - \frac{6}{x^3} - 7.$$

3. Differentiate again

$$\frac{d^2y}{dx^2} = 30x + 18x^{-4} = 30x + \frac{18}{x^4}.$$

Final answer

$$\frac{dy}{dx} = 15x^2 - \frac{6}{x^3} - 7, \text{ and } \frac{d^2y}{dx^2} = 30x + \frac{18}{x^4}.$$

Question 2

Indices & Surds

2. Given that

$$a = \frac{1}{64}x^2 \qquad b = \frac{16}{\sqrt{x}}$$

express each of the following in the form kx^n where k and n are simplified constants.

(a) $a^{\frac{1}{2}}$ (1)

(b) $\frac{16}{b^3}$ (1)

(c) $\left(\frac{ab}{2}\right)^{-\frac{4}{3}}$ (2)

(Total for Question 2 is 4 marks)

Worked Solution - Question 2

Topic group

1. Square root a

$$a^{1/2} = \left(\frac{1}{64} x^2 \right)^{1/2} = \frac{1}{8} x.$$

2. Cube b

$$b = 16x^{-1/2}, \text{ so } b^3 = 4096x^{-3/2}.$$

3. Find 16 divided by b cubed

$$\frac{16}{b^3} = \frac{16}{4096x^{-3/2}} = \frac{1}{256} x^{3/2}.$$

4. Simplify ab over 2

$$\frac{ab}{2} = \frac{1}{2} \left(\frac{1}{64} x^2 \right) (16x^{-1/2}) = \frac{1}{8} x^{3/2}.$$

5. Raise to the power -4 over 3

$$\left(\frac{1}{8} x^{3/2} \right)^{-4/3} = 8^{4/3} x^{-2} = 16x^{-2}.$$

Final answer

(a) $\frac{1}{8} x.$

(b) $\frac{1}{256} x^{3/2}.$

(c) $16x^{-2}.$

Question 3

Indices & Surds

3.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

- (a) Write $\frac{8 - \sqrt{15}}{2\sqrt{3} + \sqrt{5}}$ in the form $a\sqrt{3} + b\sqrt{5}$ where a and b are integers to be found. (3)

- (b) Hence, or otherwise, solve

$$(x + 5\sqrt{3})\sqrt{5} = 40 - 2x\sqrt{3}$$

giving your answer in simplest form.

(3)

(Total for Question 3 is 6 marks)

Worked Solution - Question 3

1. Rationalise the denominator

$$\frac{8 - \sqrt{15}}{2\sqrt{3} + \sqrt{5}} \times \frac{2\sqrt{3} - \sqrt{5}}{2\sqrt{3} - \sqrt{5}} = \frac{(8 - \sqrt{15})(2\sqrt{3} - \sqrt{5})}{12 - 5}.$$

2. Simplify the numerator

$$(8 - \sqrt{15})(2\sqrt{3} - \sqrt{5}) = 16\sqrt{3} - 8\sqrt{5} - 3\sqrt{5} + 5\sqrt{3} = 21\sqrt{3} - 14\sqrt{5}.$$

3. Divide by 7

$$\frac{21\sqrt{3} - 14\sqrt{5}}{7} = 3\sqrt{3} - 2\sqrt{5}.$$

4. Rearrange the equation

$$(x + 5\sqrt{3})\sqrt{5} = 40 - 2x\sqrt{3} \text{ gives } x(\sqrt{5} + 2\sqrt{3}) = 40 - 5\sqrt{15}.$$

5. Use the result from part (a)

$$x = \frac{40 - 5\sqrt{15}}{2\sqrt{3} + \sqrt{5}} = 5(3\sqrt{3} - 2\sqrt{5}).$$

Final answer

(a) $3\sqrt{3} - 2\sqrt{5}$.

(b) $x = 15\sqrt{3} - 10\sqrt{5}$.

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7 marks

Question 4

Graphs of Functions

4.

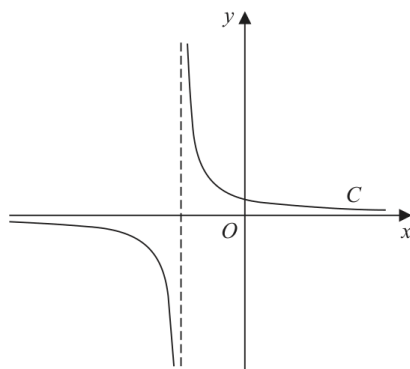


Figure 1

Figure 1 shows a sketch of part of the curve C with equation $y = \frac{1}{x+2}$

(a) State the equation of the asymptote of C that is parallel to the y -axis.

(1)

(b) Factorise fully $x^3 + 4x^2 + 4x$

(2)

A copy of Figure 1, labelled Diagram 1, is shown on the next page.

(c) On Diagram 1, add a sketch of the curve with equation

$$y = x^3 + 4x^2 + 4x$$

On your sketch, state clearly the coordinates of each point where this curve cuts or meets the coordinate axes.

(3)

(d) Hence state the number of real solutions of the equation

$$(x+2)(x^3 + 4x^2 + 4x) = 1$$

giving a reason for your answer.

(1)

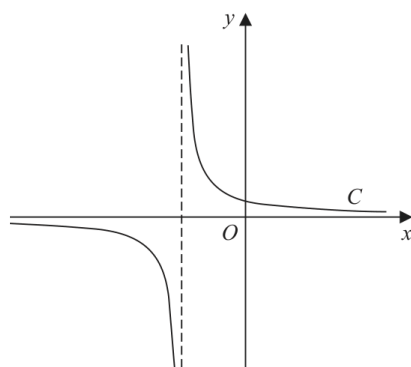
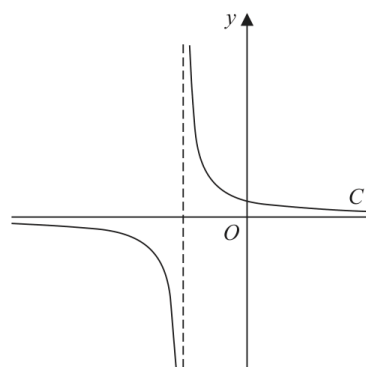


Diagram 1



copy of Diagram 1

Only use the copy of Diagram 1 if you need to redraw your answer to part (c).

(Total for Question 4 is 7 marks)

Worked Solution - Question 4

Topic group

1. Find the vertical asymptote

For $y = \frac{1}{x+2}$, the denominator is zero when $x = -2$, so the vertical asymptote is $x = -2$.

2. Factorise the cubic

$$x^3 + 4x^2 + 4x = x(x^2 + 4x + 4) = x(x+2)^2.$$

3. Use the factorised form for the sketch

$x = 0$ is a single root, so the curve crosses the x-axis at $(0, 0)$. $x = -2$ is a repeated root, so the curve touches the x-axis at $(-2, 0)$.

4. Connect the equation to intersections

$$(x+2)(x^3 + 4x^2 + 4x) = 1 \text{ is equivalent to } x^3 + 4x^2 + 4x = \frac{1}{x+2}.$$

5. Count the intersections

The number of real solutions is the number of intersections of the cubic $y = x(x+2)^2$ and the reciprocal curve $y = \frac{1}{x+2}$. The sketches show 2 intersections.

Final answer

(a) $x = -2$.

(b) $x(x+2)^2$.

(c) roots at $(-2, 0)$ and $(0, 0)$. $x = -2$ is a repeated root.

(d) 2 real roots.

Question 5

Basic Trigonometry

5.

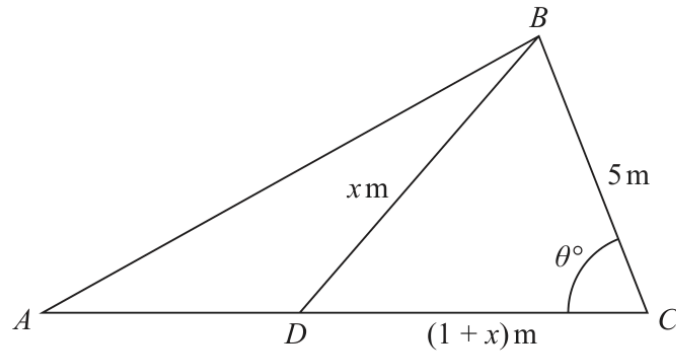
Diagram **NO**
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Figure 2

Figure 2 shows the plan view of a frame for a flat roof.

The shape of the frame consists of triangle ABD joined to triangle BCD .

Given that

- $BD = x \text{ m}$
- $CD = (1 + x) \text{ m}$
- $BC = 5 \text{ m}$
- angle $BCD = \theta^\circ$

(a) show that $\cos \theta^\circ = \frac{13 + x}{5 + 5x}$

(2)

Given also that

- $x = 2\sqrt{3}$
- angle $BAC = 30^\circ$
- ADC is a straight line

(b) find the area of triangle ABC , giving your answer, in m^2 , to one decimal place.

(5)

(Total for Question 5 is 7 marks)

Worked Solution - Question 5

Topic group

1. Apply the cosine rule in triangle BCD

$$x^2 = 5^2 + (1+x)^2 - 2(5)(1+x) \cos \theta.$$

2. Rearrange for cos theta

$$10(1+x) \cos \theta = 25 + (1+x)^2 - x^2 = 26 + 2x, \text{ so}$$

$$\cos \theta = \frac{26 + 2x}{10(1+x)} = \frac{13 + x}{5 + 5x}.$$

3. Substitute x equals 2 root 3

$$\cos C = \frac{13 + 2\sqrt{3}}{5 + 10\sqrt{3}}, \text{ so } C \approx 42.47^\circ.$$

4. Use angle sum

$$A = 30^\circ, \text{ so } B = 180^\circ - 30^\circ - C \approx 107.53^\circ.$$

5. Use the sine rule for the area

Since $BC = 5$ and $A = 30^\circ$, $\frac{BC}{\sin A} = 10$. Hence $AB = 10 \sin C$ and $AC = 10 \sin B$.

6. Calculate the area

$$\text{Area} = \frac{1}{2}(AB)(AC) \sin A = 50 \sin C \sin B \approx 32.2 \text{ m}^2.$$

Final answer

$$(a) \cos \theta = \frac{13 + x}{5 + 5x}.$$

$$(b) \text{Area} = 32.2 \text{ m}^2.$$

Question 6

Quadratics

6.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The equation

$$4(p - 2x) = \frac{12 + 15p}{x + p} \quad x \neq -p$$

where p is a constant, has two distinct real roots.

(a) Show that

$$3p^2 - 10p - 8 > 0 \quad (3)$$

(b) Hence, using algebra, find the range of possible values of p (3)

(Total for Question 6 is 6 marks)

Worked Solution - Question 6

1. Clear the denominator

$$4(p - 2x) = \frac{12 + 15p}{x + p} \text{ gives } 4(p - 2x)(x + p) = 12 + 15p.$$

2. Form a quadratic in x

Expanding gives $4p^2 - 4px - 8x^2 = 12 + 15p$, so
 $8x^2 + 4px + 12 + 15p - 4p^2 = 0$.

3. Use the discriminant condition

For two distinct real roots, $b^2 - 4ac > 0$.

4. Simplify the inequality

$$(4p)^2 - 4(8)(12 + 15p - 4p^2) > 0, \text{ so } 144p^2 - 480p - 384 > 0.$$

5. Divide and factorise

Dividing by 48 gives $3p^2 - 10p - 8 > 0$, and $3p^2 - 10p - 8 = (3p + 2)(p - 4)$.

6. State the range

The upward-opening quadratic is positive outside its roots, so $p < -\frac{2}{3}$ or $p > 4$.

Final answer

$$p < -\frac{2}{3} \text{ or } p > 4.$$

Question 7

7. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \frac{4x^2 + 10 - 7x^{\frac{1}{2}}}{4x^{\frac{1}{2}}}$

- the point $P(4, -1)$ lies on C

(a) (i) find the value of the gradient of C at P

(ii) Hence find the equation of the normal to C at P , giving your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(4)

(b) Find $f(x)$.

(6)

(Total for Question 7 is 10 marks)

Worked Solution - Question 7

1. Simplify f prime

$$f'(x) = \frac{4x^2 + 10 - 7x^{1/2}}{4x^{1/2}} = x^{3/2} + \frac{5}{2}x^{-1/2} - \frac{7}{4}.$$

2. Find the gradient at P

$$\text{At } x = 4, f'(4) = 4^{3/2} + \frac{5}{2}(4^{-1/2}) - \frac{7}{4} = 8 + \frac{5}{4} - \frac{7}{4} = \frac{15}{2}.$$

3. Find the normal gradient

The normal gradient is the negative reciprocal, so $m_n = -\frac{2}{15}$.

4. Use point P

$$\text{Using } P(4, -1), y + 1 = -\frac{2}{15}(x - 4).$$

5. Write the normal in integer form

$$15y + 15 = -2x + 8, \text{ hence } 2x + 15y + 7 = 0.$$

6. Integrate f prime

$$f(x) = \int \left(x^{3/2} + \frac{5}{2}x^{-1/2} - \frac{7}{4} \right) dx = \frac{2}{5}x^{5/2} + 5x^{1/2} - \frac{7}{4}x + c.$$

7. Use P to find c

$$-1 = \frac{2}{5}(4^{5/2}) + 5(4^{1/2}) - \frac{7}{4}(4) + c = \frac{64}{5} + 10 - 7 + c, \text{ so } c = -\frac{84}{5}.$$

Final answer

$$(a)(i) \frac{15}{2}. \quad (a)(ii) 2x + 15y + 7 = 0.$$

$$(b) f(x) = \frac{2}{5}x^{5/2} + 5x^{1/2} - \frac{7}{4}x - \frac{84}{5}.$$

Question 8

Simultaneous Equations

8.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The curve C_1 has equation

$$xy = \frac{15}{2} - 5x \quad x \neq 0$$

The curve C_2 has equation

$$y = x^3 - \frac{7}{2}x - 5$$

(a) Show that C_1 and C_2 meet when

$$2x^4 - 7x^2 - 15 = 0 \quad (2)$$

Given that C_1 and C_2 meet at points P and Q

(b) find, using algebra, the exact distance PQ (5)

(Total for Question 8 is 7 marks)

Worked Solution - Question 8

1. Write C1 in terms of y

$$xy = \frac{15}{2} - 5x, \text{ so } y = \frac{15}{2x} - 5.$$

2. Equate the two curves

$$\frac{15}{2x} - 5 = x^3 - \frac{7}{2}x - 5.$$

3. Multiply by 2x

$$15 = 2x^4 - 7x^2, \text{ hence } 2x^4 - 7x^2 - 15 = 0.$$

4. Solve for x

Let $u = x^2$. Then $2u^2 - 7u - 15 = 0 = (2u + 3)(u - 5)$, so $x^2 = 5$ and $x = \pm\sqrt{5}$.

5. Find the y-coordinates

Using $y = \frac{15}{2x} - 5$, the points are $\left(\sqrt{5}, \frac{3\sqrt{5}}{2} - 5\right)$ and $\left(-\sqrt{5}, -\frac{3\sqrt{5}}{2} - 5\right)$.

6. Use the distance formula

$$\Delta x = 2\sqrt{5} \text{ and } \Delta y = 3\sqrt{5}, \text{ so } PQ = \sqrt{(2\sqrt{5})^2 + (3\sqrt{5})^2} = \sqrt{65}.$$

Final answer

(a) $2x^4 - 7x^2 - 15 = 0$.

(b) $PQ = \sqrt{65}$.

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7 marks

Question 9

Radians

9.

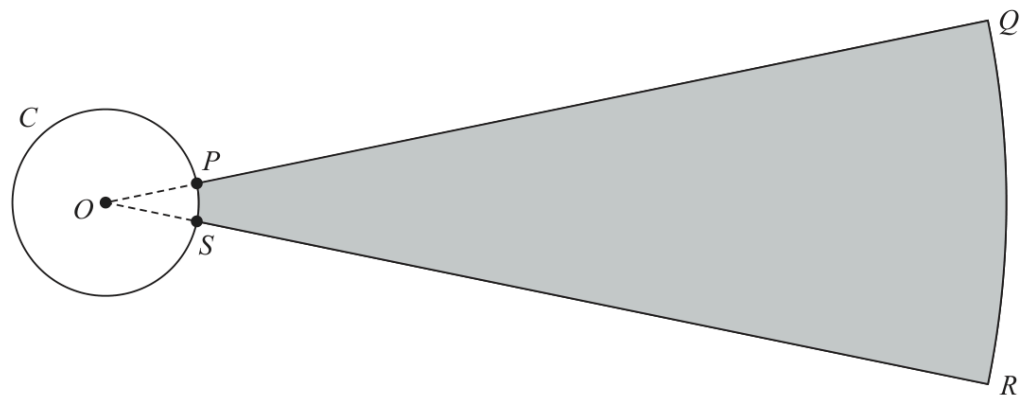
Diagram NO
accurately dr**Figure 3**

Figure 3 shows the plan view of the area being used for a ball-throwing competition.

Competitors must stand within the circle C and throw a ball as far as possible into the target area, $PQRS$, shown shaded in Figure 3.

Given that

- circle C has centre O
- P and S are points on C
- $OPQRSO$ is a sector of a circle with centre O
- the length of arc PS is 0.72 m
- the size of angle POS is 0.6 radians

(a) show that $OP = 1.2$ m

(1)

Given also that

- the target area, $PQRS$, is 90 m^2
- length $PQ = x$ metres

(b) show that

$$5x^2 + 12x - 1500 = 0$$

(3)

(c) Hence calculate the total perimeter of the target area, $PQRS$, giving your answer to the nearest metre.

(3)

(Total for Question 9 is 7 marks)

Worked Solution - Question 9

1. Use arc length

$s = r\theta$, so $0.72 = OP \times 0.6$. Hence $OP = 1.2$ m.

2. Set up the target area

The outer sector has radius $x + 1.2$ and the inner sector has radius 1.2 , with angle 0.6 radians.

3. Use sector area

$$\frac{1}{2}(x + 1.2)^2(0.6) - \frac{1}{2}(1.2)^2(0.6) = 90.$$

4. Simplify to the given quadratic

$0.3((x + 1.2)^2 - 1.44) = 90$, so $0.3x^2 + 0.72x - 90 = 0$. Multiplying by $\frac{50}{3}$ gives $5x^2 + 12x - 1500 = 0$.

5. Find the positive value of x

$$x = \frac{-12 + \sqrt{12^2 + 4(5)(1500)}}{10} = \frac{-6 + 4\sqrt{471}}{5} \approx 16.16.$$

6. Find the perimeter

Perimeter = $PQ + SR + \text{arc } QR + \text{arc } PS = x + x + 0.6(x + 1.2) + 0.72 \approx 43$ m.

7. Round to the nearest metre

The total perimeter is **43 m** to the nearest metre.

Final answer

(a) $OP = 1.2$ m.

(b) $5x^2 + 12x - 1500 = 0$.

(c) 43 m.

Question 10

Trigonometric Functions

10.

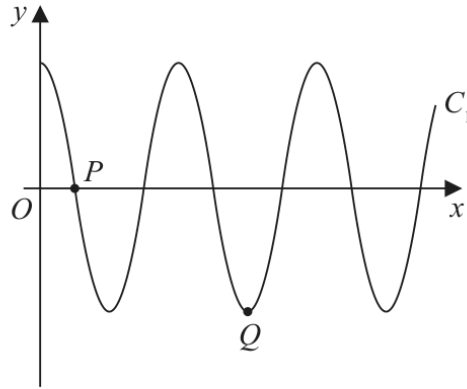


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 3 \cos\left(\frac{x}{n}\right)^\circ \quad x \geq 0$$

where n is a constant.

The curve C_1 cuts the positive x -axis for the first time at point $P(270, 0)$, as shown in Figure 4.

(a) (i) State the value of n

(ii) State the period of C_1

(2)

The point Q , shown in Figure 4, is a minimum point of C_1

(b) State the coordinates of Q .

(2)

The curve C_2 has equation $y = 2 \sin x^\circ + k$, where k is a constant.

The point $R\left(a, \frac{12}{5}\right)$ and the point $S\left(-a, -\frac{3}{5}\right)$, both lie on C_2

Given that a is a constant less than 90

(c) find the value of k .

(2)

(Total for Question 10 is 6 marks)

Worked Solution - Question 10

1. Use the first positive x-intercept

For $y = 3 \cos \left(\frac{x}{n} \right)^\circ$, the first positive zero occurs when $\frac{x}{n} = 90^\circ$.

2. Find n

Since $P = (270, 0)$, $\frac{270}{n} = 90$, so $n = 3$.

3. Find the period

The period is $360n = 360(3) = 1080^\circ$.

4. Read the shown minimum point

A minimum occurs when $\frac{x}{3} = 180^\circ + 360^\circ k$. The point Q shown is the next displayed minimum after one full period, so $x = 1620$ and $y = -3$.

5. Use the two points on C2

For $y = 2 \sin x^\circ + k$, the points give $2 \sin a + k = \frac{12}{5}$ and $-2 \sin a + k = -\frac{3}{5}$.

6. Add the equations

Adding gives $2k = \frac{12}{5} - \frac{3}{5} = \frac{9}{5}$, so $k = \frac{9}{10}$.

Final answer

(a)(i) $n = 3$. (a)(ii) 1080° .

(b) $Q = (1620, -3)$.

(c) $k = \frac{9}{10}$.

Question 11

11.

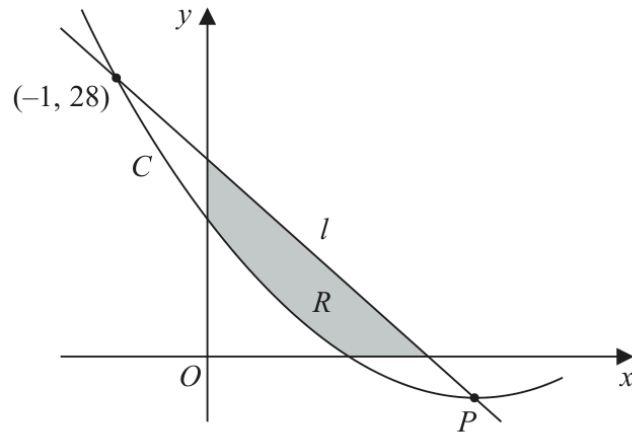


Figure 5

Figure 5 shows part of the curve C with equation $y = f(x)$ where

$$f(x) = 2x^2 - 12x + 14$$

- (a) Write $2x^2 - 12x + 14$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

Given that C has a minimum at the point P

- (b) state the coordinates of P

(1)

The line l intersects C at $(-1, 28)$ and at P as shown in Figure 5.

- (c) Find the equation of l giving your answer in the form $y = mx + c$ where m and c are constants to be found.

(3)

The finite region R , shown shaded in Figure 5, is bounded by the x -axis, l , the y -axis, and C .

- (d) Use inequalities to define the region R .

(3)

(Total for Question 11 is 10 marks)

Worked Solution - Question 11

1. Complete the square

$$2x^2 - 12x + 14 = 2(x^2 - 6x) + 14 = 2((x - 3)^2 - 9) + 14 = 2(x - 3)^2 - 4$$

2. State the minimum point

The minimum of $2(x - 3)^2 - 4$ is at $P = (3, -4)$.

3. Find the gradient of l

The line passes through $(-1, 28)$ and $P(3, -4)$, so

$$m = \frac{-4 - 28}{3 - (-1)} = \frac{-32}{4} = -8.$$

4. Find the equation of l

Using $P(3, -4)$, $y + 4 = -8(x - 3)$, so $y = -8x + 20$.

5. Use the axes as boundaries

The region is to the right of the y-axis and above the x-axis, so $x \geq 0$ and $y \geq 0$.

6. Use the line and curve as boundaries

The region is below the line and above the curve, so $y \leq -8x + 20$ and $y \geq 2x^2 - 12x + 14$.

Final answer

(a) $2(x - 3)^2 - 4$.

(b) $P = (3, -4)$.

(c) $y = -8x + 20$.

(d) $x \geq 0, y \geq 0, y \leq -8x + 20, y \geq 2x^2 - 12x + 14$.